BT504 Midterm Past paper

1 The complete set of DNAs found in each cell is known as the_____ **genome**.

2. Mouse and human genomes each seem to contain_____ protein-coding genes. **23,000**

3. Gene families originating from rare_____ **gene duplication**

4. Intervening sequences within a gene that are not translated into a protein sequence are_____ **Introns**

5. IS are found only in_____ **prokaryotes**

6. Any two humans are more than_____ identical at genetic level. **99%**

7. Yeast genome was completely sequence by_____ **1996**

8. The resolution of genetic map of genome is_____ **high**

9. Ensemble is a joint project between _____organizations. **3**

10. RED shows _____ **the diseased/cancerous sample hybridized more than the non diseased sample**

11. We use of high-throughput _____ to analyze the metabolic component of cell. **Mass Spectrometry**

12. Gene-dense areas are predominantly composed of DNA building blocks_____. **G & C**

13. Exposure to _____ breaks the DNA in human cell. **X rays**

14. miRNA Approximately_____ nucleotides long. **19-22**

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Subjective:

1. What are histones? **(2)**

Histones are a family of basic proteins that associate with DNA in the nucleus and help condense it into chromatin. Nuclear DNA does not appear in free linear strands; it is highly condensed and wrapped around histones in order to fit inside of the nucleus and take part in the formation of chromosomes.

2. Write the names of gene-dense and gene-poor areas DNA building blocks? **(2)**

Human genome's gene-dense areas are predominantly composed of the DNA building blocks G and C. Gene poor areas are rich in the DNA building blocks A and T.

3. Write three major Physical mapping techniques. **(3)**

• Cytogenetic map

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• A restriction map

• Fluorescent in situ hybridization

• Sequence tagged site (STS) map

• Nucleotide sequence map

4. Describe the D loop of Human mitochondria? (3)

The D loop is also the site where most of replication and transcription is controlled. Genes are tightly packed, with almost no non-coding DNA outside of D loop. Human mitochondrial genes contain no introns, although introns are found in the mitochondria of other groups (plants).

5. What is viral genome and write name of its types of viral genome. (5)

A viral genome is the genetic material of the virus. It is also termed the viral chromosome. Viral genomes vary in size -few thousand to more than a hundred thousand nucleotides.

• Viral genomes can be

• ssRNA

• dsRNA

• ssDNA

• dsDNA

• Linear

• Ciruclar

6. Describe a brief procedure to create labeled cDNA in Microarray. (5)

Chips are prepared by using cDNA- cDNA chips or cDNA microarray. The cDNAs are amplified. Then these immobilized on a solid support made up of nylon filter of glass slide. The probe DNA is loaded by capillary action. Small volume of this DNA is spotted on solid surface. DNA is delivered mechanically or in a robotic manner. When one DNA spotting is done, the pin is washed and loaded with fresh DNA to start the second cycle.

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